Horizon Financial Group



DATA ANALYTICS REPORT

LOAN DEFAULT RISK ANALYSIS

Credit Portfolio Overview and Risk Factor Analysis

2024 – 2025

Horizon Financial Group

DATA EXPLORATION
SQL

STATISTICAL ANALYSIS
Python

DATA VISUALIZATION
Power BI

BUSINESS INSIGHTS
Risk Recommendations

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TOOLS USED

SQL Python Power BI

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Problem Statement

Problem

Horizon Financial Group is facing elevated credit risk in its personal loan portfolio. Across 601 loan records, 146 loans were classified as defaults, resulting in an overall default rate of 24.29%. This exceeds the assumed business target of 12% by 12.29 percentage points, highlighting the need for a more data-driven approach to credit risk monitoring and loan assessment.

Highlight / Key Challenge

The main challenge is to identify borrower and loan characteristics associated with higher default rates. Factors such as credit score, debt-to-income (DTI) ratio, employment tenure, and loan purpose need to be examined to understand how default patterns differ across borrower segments. The analysis must also distinguish statistically significant relationships from differences that may occur by chance and translate the findings into practical risk-screening recommendations without relying on a single indicator for automatic loan rejection.

Analysis Objective

This analysis aims to evaluate overall portfolio performance by measuring loan volume, default count, and default rate against the assumed business target. It also seeks to identify default patterns across credit score bands, DTI categories, employment tenure groups, and loan purposes, and to assess selected relationships using correlation analysis and hypothesis testing. Finally, the analysis evaluates DTI-based screening scenarios to identify potential triggers for enhanced credit review and presents the findings through an interactive Power BI dashboard to support data-informed credit risk monitoring and decision-making.

Workflow of Analysis

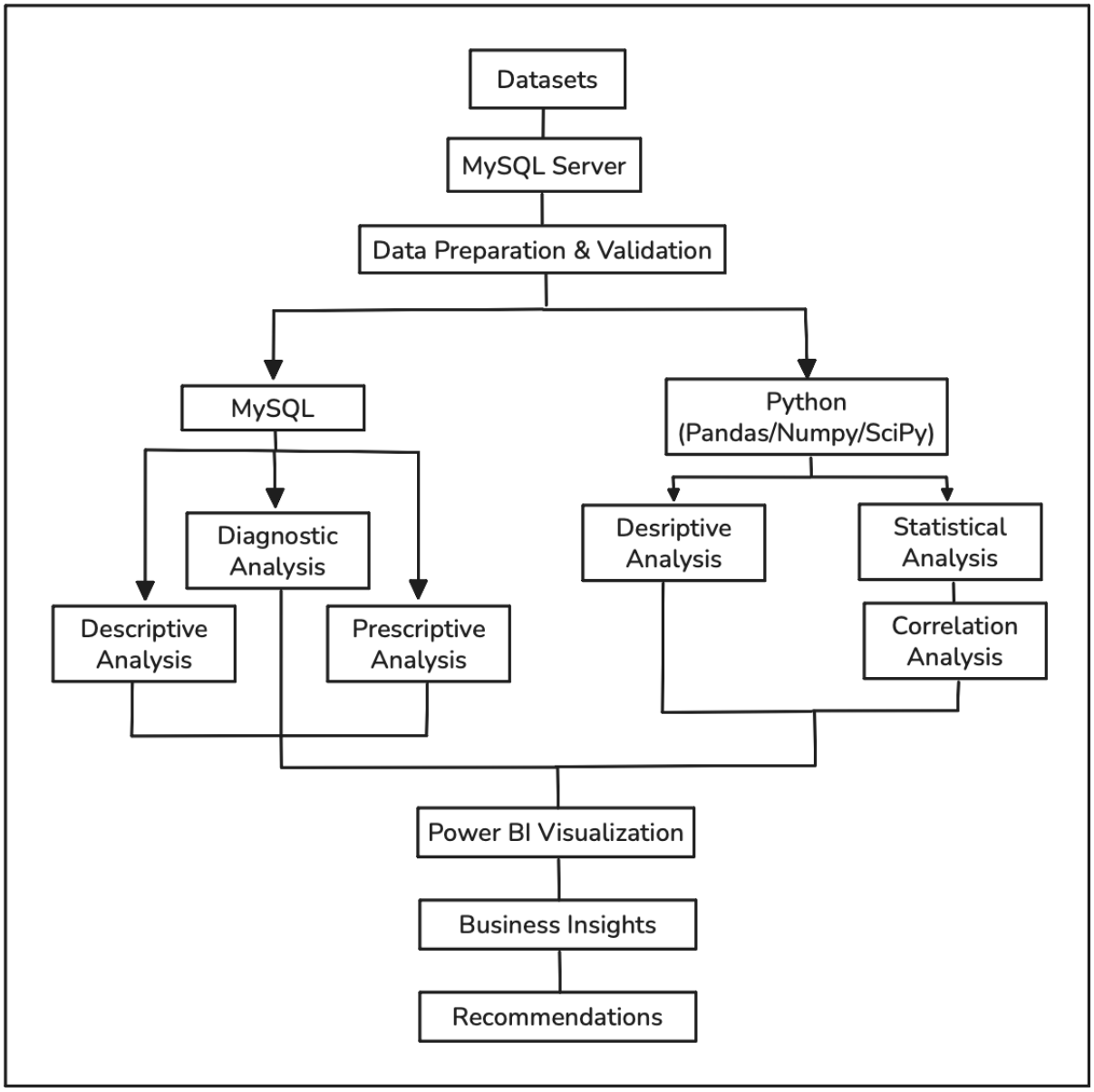


Figure 1. Workflow of Analysis

Dataset Overview

borrower_id	age	state	education_level	employment_status	years_employed	annual_income	credit_score	home_ownership	dependents	existing_monthly_debt
BRW-0001	62	CT	High School	Self-Employed	6	31020	596	Rent	1	892
BRW-0002	24	CO	High School	Full-Time	4	40188	543	Own	2	869
BRW-0003	50	NC	Bachelor	Contract	22	41014	611	Mortgage	0	1806
BRW-0004	43	TX	High School	Full-Time	19	35036	701	Mortgage	2	928
BRW-0005	46	NY	High School	Full-Time	18	44528	715	Own	0	608
BRW-0006	40	OR	Doctorate	Contract	8	157047	749	Mortgage	1	3076
BRW-0007	35	GA	Bachelor	Self-Employed	5	84210	576	Mortgage	0	1682
BRW-0008	62	KY	Bachelor	Full-Time	14	61289	563	Rent	1	939
BRW-0009	35	UT	Master	Part-Time	15	63908	643	Mortgage	2	646
BRW-0010	30	WA	Associate	Part-Time	6	22845	824	Rent	1	1044
BRW-0011	30	KY	Bachelor	Full-Time	2	43606	586	Mortgage	2	1265
BRW-0012	46	FL	Bachelor	Retired	0	44614	813	Rent	2	1184
BRW-0013	39	AL	Master	Full-Time	14	79782	615	Own	3	1207
BRW-0014	54	IN	Master	Full-Time	19	63730	850	Rent	2	671
BRW-0015	32	CA	Bachelor	Self-Employed	9	74436	535	Mortgage	0	1435
BRW-0016	41	OR	Associate	Full-Time	2	55611	570	Rent	0	1994
BRW-0017	30	AZ	High School	Full-Time	8	41753	665	Rent	4	1790
BRW-0018	68	MA	Bachelor	Self-Employed	28	66055	726	Rent	0	893
BRW-0019	43	TX	High School	Part-Time	2	23712	528	Rent	0	993
BRW-0020	43	UT	High School	Full-Time	17	45948	639	Own	3	1294
BRW-0021	37	AZ	Master	Contract	13	70937	584	Mortgage	1	2450

Figure 2. Dataset borrower_profiles.csv

Dataset information:

1. The dataset borrower_profiles.csv consist of 500 rows
2. With the borrower_id has unique ID for each row
3. Dataset already cleaned

loan_id	borrower_id	application_date	loan_purpose	loan_amount	term_months	interest_rate	monthly_payment	dti_ratio	loan_status	days_delinquent	defaulted
LN-00001	BRW-0001	2024-01-08	Vacation	16100.0	36	13.05	542.86	55.5	Paid Off	0	0
LN-00002	BRW-0001	2024-08-13	Major Purchase	5600.0	48	14.21	153.62	40.4	Default	138	1
LN-00003	BRW-0002	2025-06-28	Moving	44700.0	36	12.37	1492.59	70.5	Paid Off	0	0
LN-00004	BRW-0003	2025-05-18	Auto Loan	9200.0	48	11.18	238.58	59.8	Default	34	1
LN-00005	BRW-0004	2025-03-24	Moving	11200.0	36	9.66	359.61	44.1	Current	0	0
LN-00006	BRW-0005	2025-04-19	Education	28600.0	48	13.07	768.26	37.1	Current	0	0
LN-00007	BRW-0006	2024-07-04	Medical Expenses	39200.0	12	10.23	3450.5	49.9	Paid Off	0	0
LN-00008	BRW-0007	2025-03-29	Home Improvement	36300.0	48	14.26	996.69	38.2	Default	38	1
LN-00009	BRW-0007	2024-08-01	Business Loan	39000.0	60	13.73	902.01	36.8	Current	0	0
LN-00010	BRW-0008	2024-11-01	Debt Consolidation	28400.0	12	11.88	2521.71	67.8	Default	43	1
LN-00011	BRW-0009	2024-06-20	Business Loan	13300.0	48	10.91	343.16	18.6	Current	0	0
LN-00012	BRW-0009	2024-06-07	Business Loan	2700.0	48	12.73	72.07	13.5	Current	0	0
LN-00013	BRW-0010	2024-06-06	Wedding	15700.0	12	6.57	1355.36	126.0	Current	0	0
LN-00014	BRW-0011	2024-01-22	Moving	38300.0	48	13.9	1044.68	63.6	Current	0	0
LN-00015	BRW-0012	2024-07-13	Vacation	28800.0	60	9.94	611.06	48.3	Current	0	0
LN-00016	BRW-0012	2025-06-27	Vacation	23700.0	60	7.25	472.09	44.5	Default	105	1
LN-00017	BRW-0013	2024-07-08	Moving	2000.0	60	11.65	44.14	18.8	Current	0	0
LN-00018	BRW-0013	2024-07-13	Wedding	34300.0	60	13.85	795.44	30.1	Current	0	0
LN-00019	BRW-0014	2024-03-09	Education	17500.0	12	8.45	1525.94	41.4	Default	63	1
LN-00020	BRW-0015	2025-02-01	Debt Consolidation	20600.0	12	11.24	1822.97	52.5	Paid Off	0	0
LN-00021	BRW-0016	2024-08-01	Education	23100.0	12	15.15	2086.6	88.1	Late	15	0

Figure 3. Dataset loan_applications.csv

Dataset information:

1. The dataset loan_applications.csv consist of 601 rows
2. With the loan_id has unique ID for each row
3. Has relationship/matching column “borrower_id”

4. Dataset already cleaned

Breakdown Business Questions

Descriptive Analytics

1. What is the overall default rate, and how does it break down by credit score range?

```
142 •   SELECT
143         COUNT(*) AS total_loans,
144         SUM(defaulted) AS defaulted_loans,
145         ROUND(
146             SUM(defaulted) * 100.0 / COUNT(*),
147             2
148         ) AS overall_default_rate
149     FROM loan_applications;
150
```

100% 31:138

Result Grid Filter Rows: Search Export:

total_loans	defaulted_loans	overall_default_rate
601	146	24.29

From the datasets a total of 601 loans, with 146 loans in default and an average default rate of 24.29%.

```
155 •   SELECT
156         b.employment_status,
157         ROUND(SUM(defaulted) * 100 / COUNT(*), 2) AS defaulted_rate
158     FROM borrower_profiles b
159     JOIN loan_applications l
160         ON b.borrower_id = l.borrower_id
161     GROUP BY employment_status
162     ORDER BY defaulted_rate DESC;
163
```

100% 1:152

Result Grid Filter Rows: Search Export:

employment_status	defaulted_rate
Part-Time	27.69
Self-Employed	24.76
Full-Time	23.93
Retired	23.33
Contract	22.73

The highest defaulted rate by employment status is Part-Time with 27.69%, while the lowest defaulted rate is Contract with 22.73%.

```sql
WITH cte AS(
    SELECT
        b.borrower_id,
        b.credit_score,
        CASE
            WHEN credit_score BETWEEN 520 AND 599 THEN '520-599'
            WHEN credit_score BETWEEN 600 AND 649 THEN '600-649'
            WHEN credit_score BETWEEN 650 AND 699 THEN '650-699'
            WHEN credit_score BETWEEN 700 AND 749 THEN '700-749'
            WHEN credit_score >= 750 THEN '750+'
        END AS credit_score_bucket,
        l.loan_status,
        l.defaulted
    FROM borrower_profiles b
    JOIN loan_applications l
        ON b.borrower_id = l.borrower_id
)
SELECT
    credit_score_bucket,
    COUNT(*) AS total_loans,
    SUM(defaulted) AS defaulted_loans,
    ROUND(SUM(defaulted) * 100 / COUNT(*), 2) AS defaulted_rate
FROM cte
GROUP BY credit_score_bucket
ORDER BY credit_score_bucket ASC;
```

75% 34:225

Result Grid Filter Rows: Search Export:

credit_score_bucket	total_loans	defaulted_loans	defaulted_rate
520-599	116	57	49.14
600-649	93	27	29.03
650-699	75	21	28.00
700-749	86	14	16.28
750+	231	27	11.69

The highest defaulted rate represent the lowest credit score bucket range between 520 - 599 with 49.14% defaulted rate for the detail with total 116 total_loans and 57 defaulted_loans

2. Which loan purposes have the highest default rates?

```sql
SELECT
    l.loan_purpose,
    ROUND(SUM(defaulted) * 100 / COUNT(*), 2) AS defaulted_rate
FROM borrower_profiles b
JOIN loan_applications l
    ON b.borrower_id = l.borrower_id
GROUP BY loan_purpose
ORDER BY defaulted_rate DESC;
```

5% 57:230

Result Grid Filter Rows: Search Export:

loan_purpose	defaulted_rate
Wedding	32.14
Home Improvement	28.57
Auto Loan	27.12
Business Loan	24.14
Education	22.64
Vacation	22.58
Major Purchase	22.06
Debt Consolidation	21.57
Moving	21.43
Medical Expenses	20.59

The highest defaulted rate by loan purpose is Wedding with 32.14%, while the lowest defaulted rate is Medical Expenses with 20.59%.

3. How do employment status and years employed affect default risk?

```
248    SELECT
249        b.employment_status,
250        COUNT(*) AS total_loans,
251        SUM(l.defaulted) AS defaulted_loans,
252        ROUND(SUM(l.defaulted) * 100.0 / COUNT(*), 2) AS default_rate
253    FROM borrower_profiles b
254    JOIN loan_applications l
255        ON b.borrower_id = l.borrower_id
256    GROUP BY b.employment_status
257    ORDER BY default_rate DESC;
258
```

5% 1:245

Result Grid Filter Rows: Search Export:

employment_status	total_loans	defaulted_loans	default_rate
Part-Time	65	18	27.69
Self-Employed	105	26	24.76
Full-Time	305	73	23.93
Retired	60	14	23.33
Contract	66	15	22.73

The highest defaulted rate by employment_status is Part-Time with 27.69% defaulted rate, while the lowest defaulted rate by employment_status is Contract with 22.73% defaulted rate.

```sql
WITH cte AS (
    SELECT
        b.borrower_id,
        b.years_employed,
        l.defaulted,
        CASE
            WHEN b.years_employed < 2 THEN '<2 years'
            WHEN b.years_employed BETWEEN 2 AND 5 THEN '2-5 years'
            WHEN b.years_employed BETWEEN 6 AND 10 THEN '6-10 years'
            WHEN b.years_employed BETWEEN 11 AND 20 THEN '11-20 years'
            ELSE '21+ years'
        END AS years_employed_bands
    FROM borrower_profiles b
    JOIN loan_applications l
        ON b.borrower_id = l.borrower_id
)
SELECT
    years_employed_bands,
    COUNT(*) AS total_loans,
    SUM(defaulted) AS defaulted_loans,
    ROUND(SUM(defaulted) * 100.0 / COUNT(*), 2) AS default_rate
FROM cte
GROUP BY years_employed_bands
ORDER BY
    CASE years_employed_bands
        WHEN '<2 years' THEN 1
        WHEN '2-5 years' THEN 2
        WHEN '6-10 years' THEN 3
        WHEN '11-20 years' THEN 4
        WHEN '21+ years' THEN 5
    END;
```

)% 1:261

Result Grid Filter Rows: Search Export:

years_employed_ban...	total_loans	defaulted_loans	default_rate
<2 years	84	29	34.52
2-5 years	146	24	16.44
6-10 years	110	33	30.00
11-20 years	161	34	21.12
21+ years	100	26	26.00

The highest defaulted rate by years employed is less than 2 years, while the lowest defaulted rate by years employed is 2-5 years.

4. Which loan status has the most loan amount?

```
319    SELECT
320        loan_status,
321        COUNT(*) AS total_loans,
322        ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM loan_applications), 2) AS loan_share_pct,
323        ROUND(SUM(loan_amount), 2) AS total_loan_amount,
324        ROUND(AVG(loan_amount), 2) AS avg_loan_amount
325    FROM loan_applications
326    GROUP BY loan_status
327    ORDER BY total_loan_amount DESC;
328
```

25:310

Result Grid Filter Rows: Search Export:

loan_status	total_loans	loan_share_pct	total_loan_amount	avg_loan_amount
Current	239	39.77	5089300.00	21294.14
Default	146	24.29	3295300.00	22570.55
Late	103	17.14	2510500.00	24373.79
Paid Off	113	18.80	2416000.00	21380.53

The highest total loan amount by loan_status is Current with $5,089,300 total loan amount, while the lowest total loan amount by loan_status is Paid Off with $2,416,000 total loan amount.

5. What is average loan amount by defaulted vs non-defaulted?

```
334    SELECT
335        CASE
336            WHEN defaulted = 1 THEN 'Defaulted'
337            ELSE 'Non-Defaulted'
338        END AS defaulted_status,
339        COUNT(*) AS total_loans,
340        ROUND(AVG(loan_amount), 2) AS average_loan_amount,
341        ROUND(MIN(loan_amount), 2) AS minimum_loan_amount,
342        ROUND(MAX(loan_amount), 2) AS maximum_loan_amount
343    FROM loan_applications
344    GROUP BY defaulted_status;
345
```

94:329

Result Grid Filter Rows: Search Export:

defaulted_stat...	total_loans	average_loan_amount	minimum_loan_amount	maximum_loan_amount
Non-Defaulted	455	22012.75	2000.00	49800.00
Defaulted	146	22570.55	2400.00	49700.00

Average loan amount on Defaulted by defaulted_status is $22,571, while average loan amount on Non-Defaulted by defaulted status is $22,013

6. How does Loan Characteristics by total loan based on loan purpose and loan terms?

```
352 •  SELECT
353        loan_purpose,
354        COUNT(*) AS total_loans,
355        ROUND(AVG(loan_amount), 2) AS avg_loan_amount,
356        ROUND(AVG(interest_rate), 2) AS avg_interest_rate,
357        ROUND(AVG(monthly_payment), 2) AS avg_monthly_payment
358    FROM loan_applications
359    GROUP BY loan_purpose
360    ORDER BY total_loans DESC;
361
```

75% 1:348

Result Grid Filter Rows: Search Export:

loan_purpose	total_loans	avg_loan_amount	avg_interest_...	avg_monthly_paym...
Home Improvement	70	23407.14	10.71	933.32
Major Purchase	68	23164.71	10.72	1127.02
Medical Expenses	68	23176.47	10.72	1014.13
Vacation	62	24477.42	10.56	945.74
Auto Loan	59	19554.24	10.63	835.86
Business Loan	58	21013.79	10.76	843.37
Moving	56	21300.00	10.10	782.49
Wedding	56	21892.86	10.63	1029.82
Education	53	19847.17	11.45	841.88
Debt Consolidation	51	22756.86	10.08	1098.89

The highest total loans by loan_purpose is Home Improvement with 70 total_loans, while the lowest total loans by loan purpose is Debt Consolidation with 51 total_loans.

```
367 •  SELECT
368        term_months,
369        COUNT(*) AS total_loans,
370        ROUND(AVG(loan_amount), 2) AS avg_loan_amount,
371        ROUND(AVG(interest_rate), 2) AS avg_interest_rate,
372        ROUND(AVG(monthly_payment), 2) AS avg_monthly_payment
373    FROM loan_applications
374    GROUP BY term_months
375    ORDER BY term_months;
376
```

75% 22:359

Result Grid Filter Rows: Search Export:

term_months	total_loans	avg_loan_amount	avg_interest_...	avg_monthly_paym...
12	118	22978.81	10.72	2029.79
24	110	22464.55	10.75	1043.28
36	118	22311.02	10.57	726.34
48	137	20697.81	10.77	531.54
60	118	22544.07	10.38	484.28

The highest total loans by term months is 48 Months with 137 total loans, while the lowest total loans by term months is 24 Months with 110 total loans.

Diagnostics Analytics

1. What factors may explain higher default rates among certain loan purposes?

```
384     -- Identify the problem is
385     SELECT
386         loan_purpose,
387         COUNT(*) AS total_loans,
388         SUM(defaulted) AS defaulted_loans,
389         ROUND(
390             SUM(defaulted) * 100.0 / COUNT(*),
391             2
392         ) AS default_rate
393     FROM loan_applications
394     GROUP BY loan_purpose
395     ORDER BY default_rate DESC;
396
```

75% 1:380

Result Grid Filter Rows: Search Export:

loan_purpose	total_loans	defaulted_loans	default_rate
Wedding	56	18	32.14
Home Improvement	70	20	28.57
Auto Loan	59	16	27.12
Business Loan	58	14	24.14
Education	53	12	22.64
Vacation	62	14	22.58
Major Purchase	68	15	22.06
Debt Consolidation	51	11	21.57
Moving	56	12	21.43
Medical Expenses	68	14	20.59

Wedding has the highest default rate with 32.14%, while Medical Expenses has the lowest default rate with 20.59%

```sql
-- Compare characteristics in loan purpose
SELECT
    loan_purpose,
    COUNT(*) AS total_loans,
    ROUND(AVG(loan_amount), 2) AS avg_loan_amount,
    ROUND(AVG(term_months), 2) AS avg_term_months,
    ROUND(AVG(interest_rate), 2) AS avg_interest_rate,
    ROUND(AVG(monthly_payment), 2) AS avg_monthly_payment,
    ROUND(AVG(dti_ratio), 2) AS avg_dti,
    ROUND(SUM(defaulted) * 100.0 / COUNT(*), 2) AS default_rate
FROM loan_applications
GROUP BY loan_purpose
ORDER BY default_rate DESC;
```

75% 28:395

Result Grid Filter Rows: Search Export:

loan_purpose	total_loans	avg_loan_amount	avg_term_mont...	avg_interest_...	avg_monthly_paym...	avg_dti	default_rate
Wedding	56	21892.86	33.21	10.63	1029.82	52.38	32.14
Home Improvement	70	23407.14	39.09	10.71	933.32	51.19	28.57
Auto Loan	59	19554.24	37.22	10.63	835.86	44.39	27.12
Business Loan	58	21013.79	37.03	10.76	843.37	43.75	24.14
Education	53	19847.17	37.36	11.45	841.88	46.11	22.64
Vacation	62	24477.42	38.90	10.56	945.74	52.02	22.58
Major Purchase	68	23164.71	33.88	10.72	1127.02	54.34	22.06
Debt Consolidation	51	22756.86	33.41	10.08	1098.89	50.78	21.57
Moving	56	21300.00	38.14	10.10	782.49	46.52	21.43
Medical Expenses	68	23176.47	36.53	10.72	1014.13	53.32	20.59

Wedding, Home Improvement, and Auto Loan have the highest observed default rates. These purposes generally show relatively high debt burden or repayment obligations, but the relationship is not consistent across all purposes. Therefore, loan-level characteristics such as DTI, loan amount, and monthly payment may contribute to differences in default rates, but no single factor clearly explains the variation.

```sql
SELECT
    loan_purpose,
    CASE
        WHEN defaulted = 1 THEN 'Defaulted'
        ELSE 'Non-Defaulted'
    END AS default_status,
    COUNT(*) AS total_loans,
    ROUND(AVG(loan_amount), 2) AS avg_loan_amount,
    ROUND(AVG(term_months), 2) AS avg_term_months,
    ROUND(AVG(interest_rate), 2) AS avg_interest_rate,
    ROUND(AVG(monthly_payment), 2) AS avg_monthly_payment,
    ROUND(AVG(dti_ratio), 2) AS avg_dti
FROM loan_applications
GROUP BY
    loan_purpose,
    defaulted
ORDER BY
    loan_purpose,
    default_status;
```

75% 13:420

Result Grid Filter Rows: Search Export:

loan_purpose	default_status	total_loans	avg_loan_amount	avg_term_mont...	avg_interest_...	avg_monthly_paym...	avg_dti
Business Loan	Defaulted	14	16407.14	33.43	11.40	840.31	59.01
Business Loan	Non-Defaulted	44	22479.55	38.18	10.56	844.34	38.90
Debt Consolidation	Defaulted	11	29036.36	31.64	9.76	1710.35	68.22
Debt Consolidation	Non-Defaulted	40	21030.00	33.90	10.17	930.74	45.99
Education	Defaulted	12	29183.33	34.00	12.30	1436.28	55.13
Education	Non-Defaulted	41	17114.63	38.34	11.20	667.91	43.47
Home Improvement	Defaulted	20	19150.00	40.20	11.65	737.33	54.63
Home Improvement	Non-Defaulted	50	25110.00	38.64	10.34	1011.72	49.81
Major Purchase	Defaulted	15	16180.00	30.40	12.00	833.53	57.06
Major Purchase	Non-Defaulted	53	25141.51	34.87	10.36	1210.09	53.58

Higher default rates across certain loan purposes appear to be associated with higher borrower debt burden and repayment obligations. Defaulted loans generally show higher average DTI and, in several loan-purpose categories, higher interest rates and monthly payments than non-defaulted loans. However, loan amount and loan term do not show a consistent pattern across purposes.

2. What factors may explain higher default rates among borrowers with shorter employment tenure?

```sql
SELECT
    CASE
        WHEN b.years_employed < 2 THEN '<2 years'
        ELSE '2+ years'
    END AS employment_group,
    COUNT(*) AS total_loans,
    SUM(l.defaulted) AS defaulted_loans,
    ROUND(SUM(l.defaulted) * 100.0 / COUNT(*), 2) AS default_rate,
    ROUND(AVG(b.annual_income), 2) AS avg_annual_income,
    ROUND(AVG(b.credit_score), 2) AS avg_credit_score,
    ROUND(AVG(b.existing_monthly_debt), 2) AS avg_existing_debt,
    ROUND(AVG(l.loan_amount), 2) AS avg_loan_amount,
    ROUND(AVG(l.monthly_payment), 2) AS avg_monthly_payment,
    ROUND(AVG(l.dti_ratio), 2) AS avg_dti
FROM borrower_profiles b
JOIN loan_applications l
    ON b.borrower_id = l.borrower_id
GROUP BY employment_group;
```

75% 27:500

Result Grid Filter Rows: Search Export:

employment_group	total_loans	defaulted_loans	default_rate	avg_annual_inco...	avg_credit_sc...	avg_existing_debt	avg_loan_amount	avg_monthly_paym...	avg_dti
2+ years	517	117	22.63	64725.50	703.71	1555.07	22075.44	948.60	48.64
<2 years	84	29	34.52	67093.76	718.29	2049.55	22596.43	946.75	56.07

Borrowers with less than 2 years of employment have a higher observed default rate (34.52% vs 22.63%). This group also has higher average existing monthly debt and DTI, suggesting that greater debt burden may be one factor associated with the higher default rate. However, credit score and income do not show patterns that clearly explain the difference.

Prescriptive Analytics

1. What DTI threshold would you recommend as a cutoff for loan approval?

```sql
WITH thresholds AS (
    SELECT 40 AS dti_threshold
    UNION ALL SELECT 45
    UNION ALL SELECT 50
    UNION ALL SELECT 55
    UNION ALL SELECT 60
    UNION ALL SELECT 65
    UNION ALL SELECT 70
)
SELECT
    t.dti_threshold,
    COUNT(l.loan_id) AS total_loans_above_cutoff,
    SUM(l.defaulted) AS defaults_above_cutoff,
    ROUND(SUM(l.defaulted) * 100.0 / COUNT(l.loan_id), 2) AS default_rate_above_cutoff,
    ROUND(COUNT(l.loan_id) * 100.0 / (SELECT COUNT(*) FROM loan_applications), 2) AS loan_pct_above_cutoff,
    ROUND(SUM(l.defaulted) * 100.0 / (SELECT SUM(defaulted) FROM loan_applications), 2) AS default_capture_rate
FROM thresholds t
JOIN loan_applications l
    ON l.dti_ratio > t.dti_threshold
GROUP BY t.dti_threshold
ORDER BY t.dti_threshold;
```

75% 71:532

Result Grid Filter Rows: Search Export:

dti_threshold	total_loans_above_cutoff	defaults_above_cutoff	default_rate_above_cutoff	loan_pct_above_cutoff	default_capture_r...
40	364	115	31.59	60.57	78.77
45	330	105	31.82	54.91	71.92
50	270	93	34.44	44.93	63.70
55	219	81	36.99	36.44	55.48
60	171	61	35.67	28.45	41.78
65	128	46	35.94	21.30	31.51
70	95	36	37.89	15.81	24.66

A 60% DTI threshold is recommended as an initial screening cutoff. Loans above 60% represent 28.45% of the portfolio and have a 35.67% observed default rate, while accounting for 41.78% of all observed defaults. This provides a practical balance between identifying higher-risk borrowers and maintaining reasonable loan approval coverage. The threshold should be used as a risk-screening criterion rather than an automatic rejection rule and should be combined with other borrower characteristics such as credit score and existing debt.

Deeper Analysis using Python

1. Descriptive Statistic

```
df.describe().round(2)
```

	annual_income	existing_monthly_debt	credit_score	years_employed	loan_amount	term_m
count	601.00	601.00	601.00	601.00	601.00	
mean	65056.51	1624.18	705.75	10.92	22148.25	
std	29217.56	1124.97	98.22	9.35	13215.26	
min	12549.00	124.00	528.00	0.00	2000.00	
25%	43022.00	791.00	617.00	3.00	10900.00	
50%	59967.00	1316.00	707.00	8.00	20400.00	
75%	81309.00	2207.00	796.00	17.00	32700.00	
max	157047.00	6130.00	850.00	35.00	49800.00	

The dataset contains 601 loan records. The average annual income is approximately 65,056.51, while the average loan amount is 22,148.25 and the average loan term is 36.54 months. The average DTI ratio is 49.68%, with a standard deviation of 24.11 percentage points, indicating substantial variation in borrowers' debt burden.

The observed default rate is 24.29%, exceeding the business target of 12%. Several variables, including annual income, existing monthly debt, loan amount, and monthly payment, have means higher than their medians, suggesting potential right-skewness that should be investigated further. These descriptive findings provide an initial overview of borrower and loan characteristics but do not establish which factors are statistically associated with default.

2. Correlation Analysis

Is the relationship between DTI and default statistically significant?
Metode: Point-Biserial Correlation

H0: There is no significant linear relationship between DTI ratio and loan default.
H1: There is a significant linear relationship between DTI ratio and loan default.

```
# Prepare the 2 variables dti ratio vs default

corr_df = df[["dti_ratio", "defaulted"]].dropna()

print("Total observations:", len(corr_df))
print(corr_df["defaulted"].value_counts())
```

```
Total observations: 601
defaulted
0    455
1    146
Name: count, dtype: int64
```

```
# Measure point-biserial correlation
corr, p_value = stats.pointbiserialr(
    corr_df["defaulted"],
    corr_df["dti_ratio"]
)

print(f"Correlation coefficient: {corr:.4f}")
print(f"P-value: {p_value:.4f}")
```

```
Correlation coefficient: 0.1925
P-value: 0.0000
```

The point-biserial correlation analysis identified a weak positive relationship between debt-to-income (DTI) ratio and loan default ($r = 0.1925$). The relationship was statistically significant at the 5% significance level ($p < 0.05$). This suggests that borrowers with higher DTI ratios tend to have a greater likelihood of default in the observed dataset. However, the weak correlation indicates that DTI alone does not fully explain default risk, and the result does not establish causation. Other borrower and loan characteristics should be investigated to provide a more comprehensive assessment of default risk.

Accept H1: There is a significant linear relationship between DTI ratio and loan default.

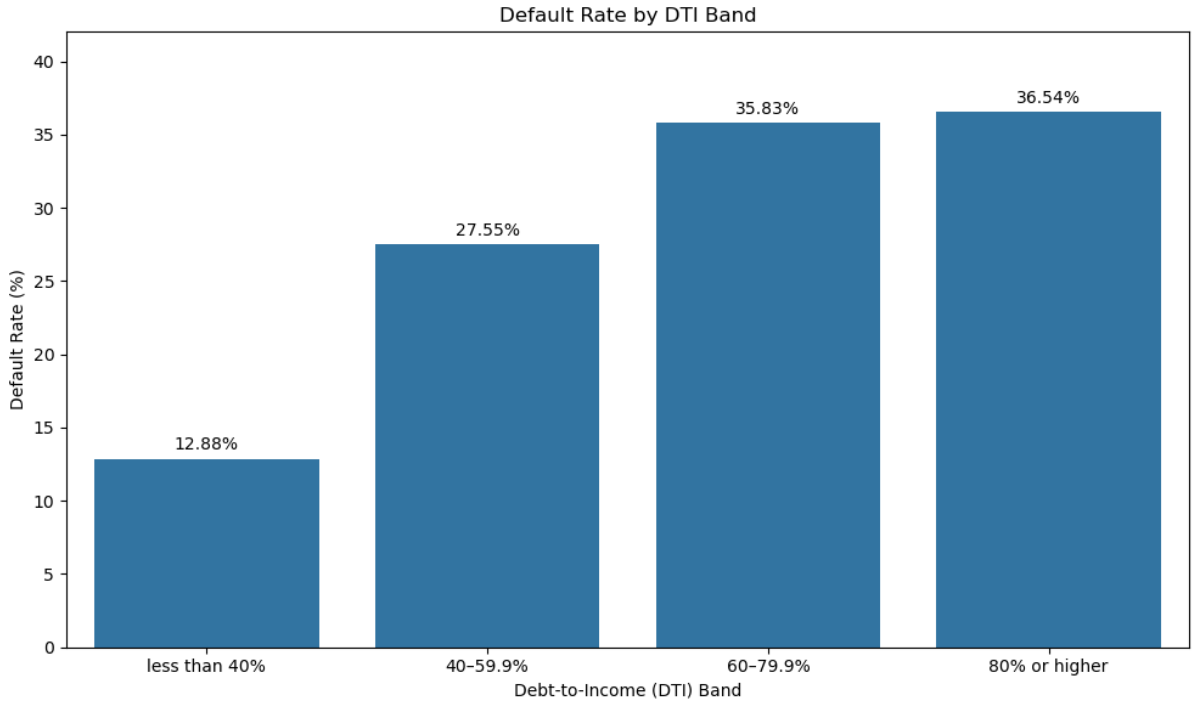


Figure 4. Default Rate by DTI Band Matplotlib Seaborn

Does the average loan amount differ significantly between defaulted and non-defaulted loans?
Metode: Welch's t-test

H0: The mean loan amount is equal for defaulted and non-defaulted loans.
H1: The mean loan amount differs between defaulted and non-defaulted loans.

```
t_stat, p_value = stats.ttest_ind(
    defaulted_loans,
    non_defaulted_loans,
    equal_var=False,
    alternative="two-sided"
)

print(f"T-statistic: {t_stat:.4f}")
print(f"P-value: {p_value:.6f}")
```

```
T-statistic: 0.4446
P-value: 0.656969
```

Welch's t-test was conducted to examine whether the average loan amount differed significantly between defaulted and non-defaulted loans. The test returned a t-statistic of 0.4446 and a p-value of 0.656969. Since the p-value exceeds the 0.05 significance level, the null hypothesis was not rejected. Therefore, the analysis found insufficient statistical evidence to conclude that average loan amounts differ significantly between the two groups. This suggests that loan amount alone may not be a sufficient indicator for distinguishing default risk in this dataset.

Accept H0: The mean loan amount is equal for defaulted and non-defaulted loans.

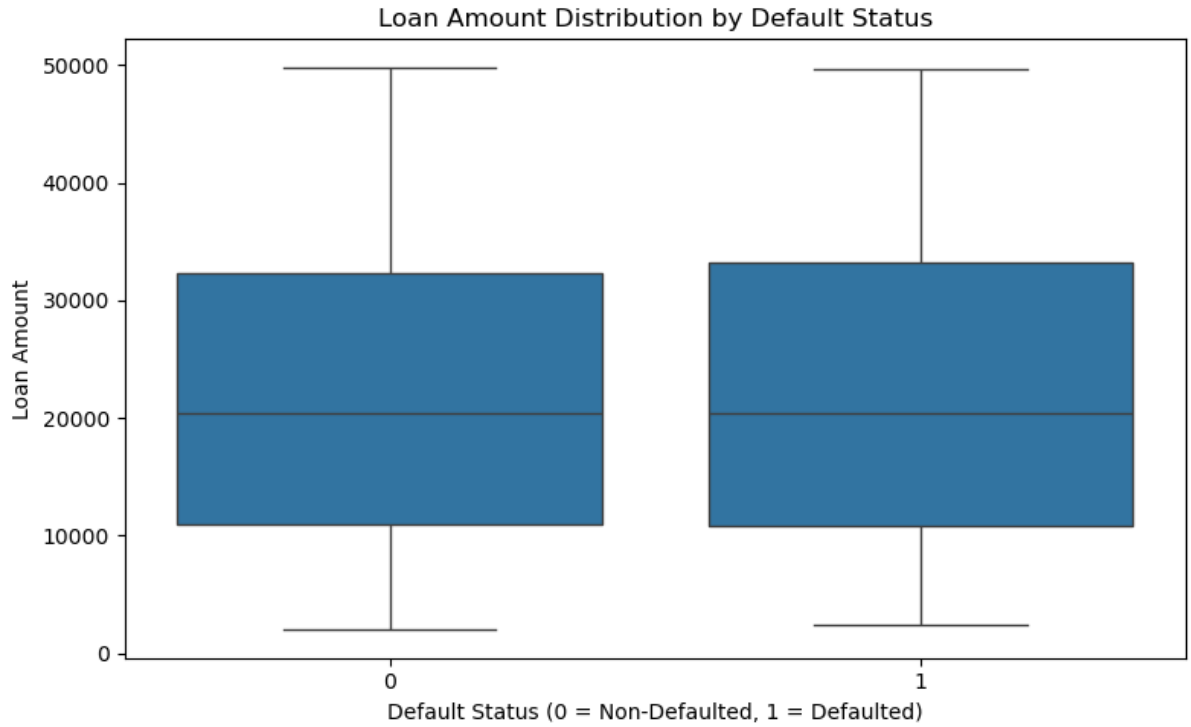


Figure 5. Loan Amount Distribution by Default Status Matplotlib Seaborn

Are default rates significantly associated with loan purpose?
Metode: Chi-square Test of Independence

H0: Loan purpose and default status are independent; there is no significant association between them.
H1: Loan purpose and default status are not independent; there is a significant association between them.

```
chi2, p_value, dof, expected = stats.chi2_contingency(
    contingency_table
)

print(f"Chi-square statistic: {chi2:.4f}")
print(f"Degrees of freedom: {dof}")
print(f"P-value: {p_value:.6f}")
```

```
Chi-square statistic: 4.1550
Degrees of freedom: 9
P-value: 0.900917
```

A Chi-square test of independence was conducted to examine the association between loan purpose and default status. The test produced a chi-square statistic of 4.1550 with 9 degrees of freedom and a p-value of 0.900917. Since the p-value exceeds the 0.05 significance level, the null hypothesis was not rejected. Therefore, the analysis found insufficient statistical evidence of an association between loan purpose and default status in this dataset. Although observed default rates varied across loan purposes, these differences were not statistically significant. The expected-frequency assumption was also satisfied, with no expected cell frequencies below 5.

Accept H0: Loan purpose and default status are independent; there is no significant association between them.

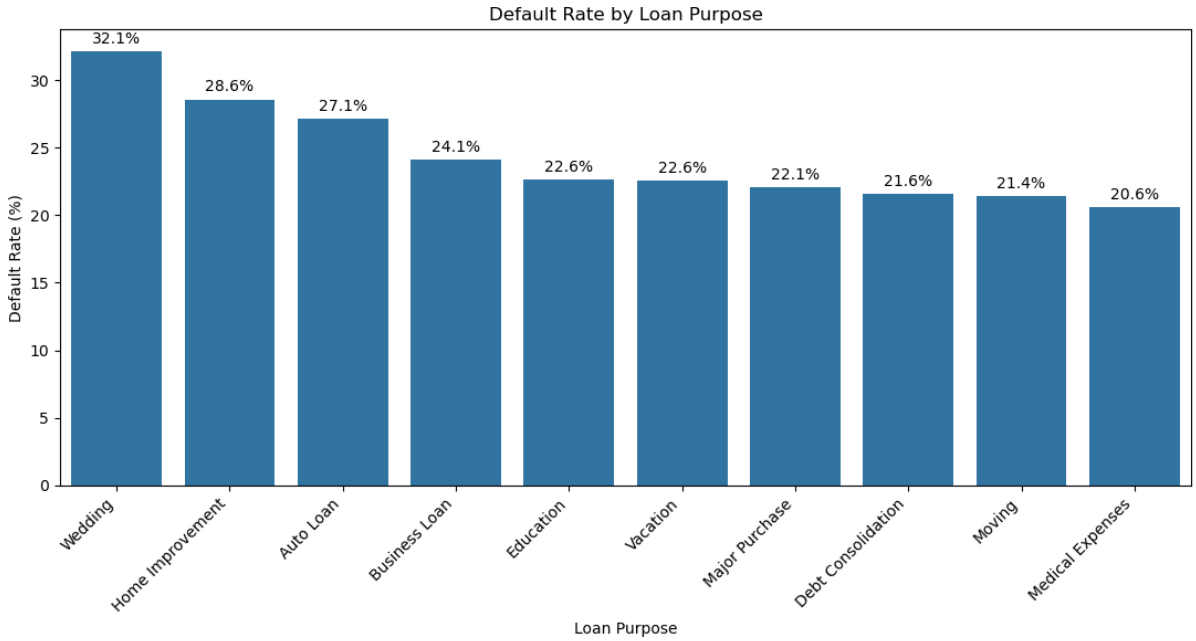


Figure 6. Default Rate by Loan Purpose Matplotlib Seaborn

Are borrowers with less than 2 years of employment significantly more likely to default?
Metode: One-sided Two-Proportion Z-Test

H0: The default proportion among borrowers with less than 2 years of employment is not higher than that among borrowers with 2 or more years of employment.
H1: The default proportion among borrowers with less than 2 years of employment is higher than that among borrowers with 2 or more years of employment.

```
z_stat, p_value = proportions_ztest(
    count=defaults,
    nobs=totals,
    alternative="larger"
)

print(f"Z-statistic: {z_stat:.4f}")
print(f"P-value: {p_value:.6f}")
```

```
Z-statistic: 2.3574
P-value: 0.009201
```

A one-sided two-proportion z-test was conducted to determine whether borrowers with less than two years of employment had a higher default rate than borrowers with two or more years of employment. The default rate was 34.52% for the shorter-tenure group (29 of 84 loans), compared with 22.63% for the longer-tenure group (117 of 517 loans), a difference of 11.89 percentage points. The test produced a z-statistic of 2.3574 and a p-value of 0.009201. Since the p-value is below the 0.05 significance level, the null hypothesis was rejected. The results provide statistical evidence that

borrowers with less than two years of employment had a higher default proportion in this dataset. However, this association does not establish causation, and other borrower characteristics should also be considered in credit risk assessment.

Accept H1: The default proportion among borrowers with less than 2 years of employment is higher than that among borrowers with 2 or more years of employment.

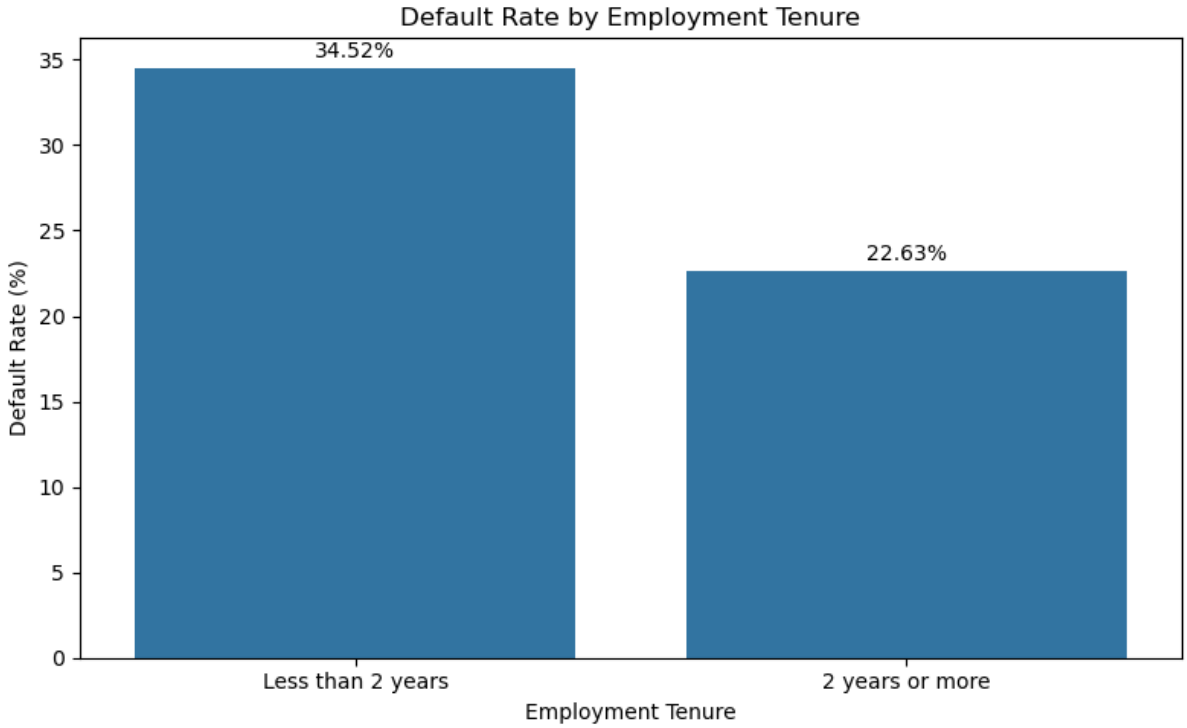
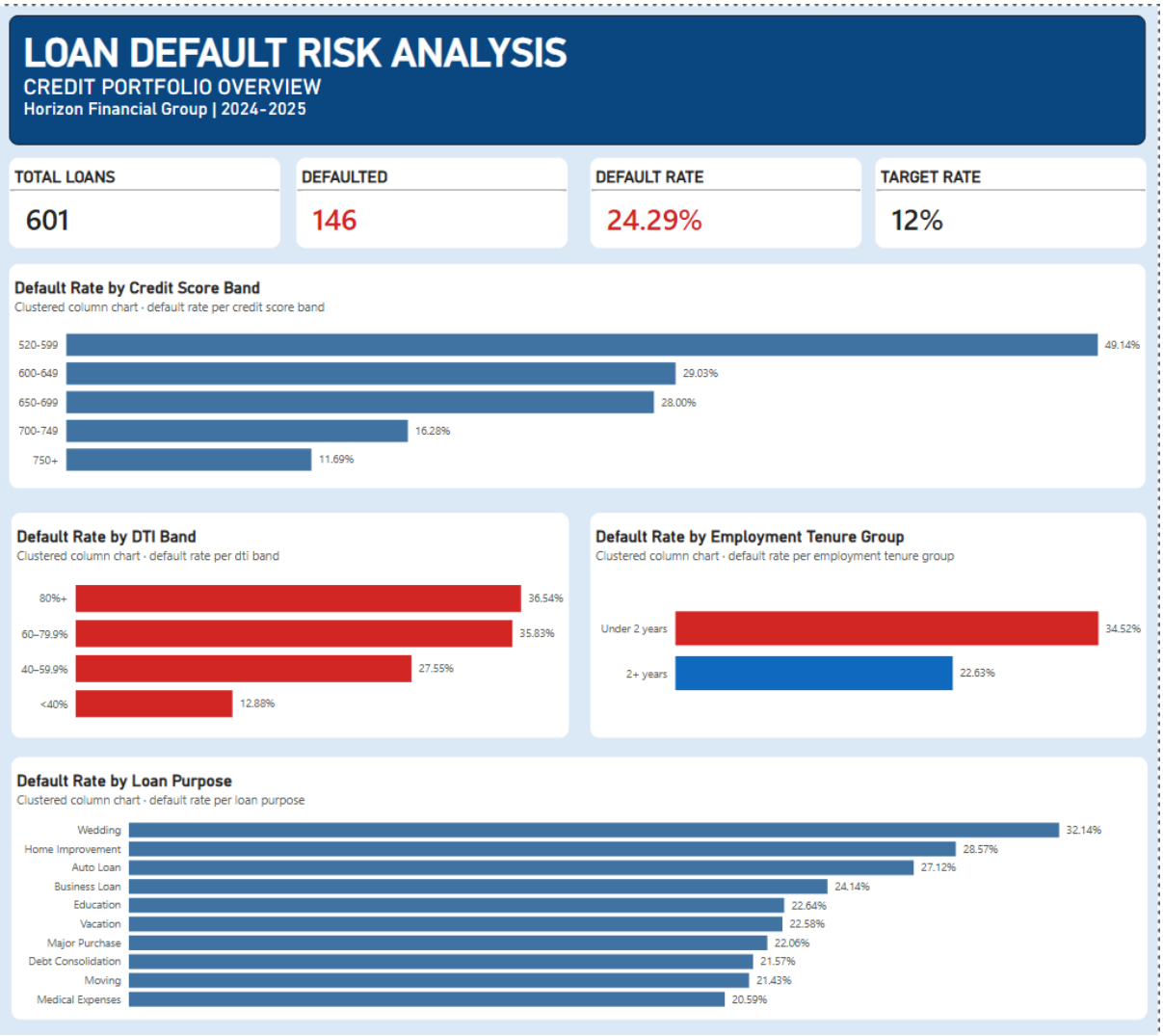


Figure 7. Default Rate by Employment Tenure

Visualization

Using Microsoft Power BI with integrated from MySQL server to build visualization dashboard and storytelling after analyzing the datas.



The dashboard reveals that the loan portfolio has an overall default rate of 24.29% across 601 loans, with 146 defaults, exceeding the assumed business target of 12% by 12.29 percentage points. Borrowers with credit scores of 520–599 recorded the highest default rate at 49.14%, while those with scores of 750 or above had the lowest rate at 11.69%, suggesting that lower credit scores are associated with greater default risk. Similarly, loans with DTI ratios of 60% or higher showed elevated default rates, particularly the 80%+ DTI group at 36.54%. Borrowers with less than two years of employment tenure also recorded a higher default rate (34.52%) than those with two or more years (22.63%). Although default rates varied by loan purpose, statistical testing found no significant overall association between loan purpose and default ($p = 0.901$). These findings suggest that credit score, repayment capacity, and employment tenure may be useful indicators for enhanced risk assessment, while loan purpose should not be used as a standalone risk indicator. The results describe observed patterns and should support further review rather than be interpreted as proof of causation.

Insights

The analysis indicates that the portfolio's default rate of 24.29% is substantially above the assumed business target of 12%. Lower credit scores are associated with higher default rates, with the 520–599 group recording the highest rate at 49.14%, while borrowers with DTI ratios of 60% or higher also show elevated default rates. In addition, borrowers with less than two years of employment tenure have a higher default rate (34.52%) than those with longer tenure (22.63%). However, loan purpose was not significantly associated with default in the chi-square test ($p = 0.901$), suggesting that observed differences across loan purposes should be interpreted cautiously.

Recommendations

Horizon Financial Group should strengthen its credit risk assessment by reviewing credit score, DTI ratio, and employment tenure together rather than relying on a single indicator. Loans with DTI ratios above 60% could be flagged for enhanced review, as an initial screening scenario, while lower credit scores and shorter employment tenure may prompt additional affordability and repayment-capacity checks. The proposed DTI threshold should be evaluated against business objectives, potential false positives, and lending policies before implementation, and should not automatically trigger loan rejection. The organization should also monitor default rates regularly, validate screening rules on additional data, and use the Power BI dashboard to track portfolio performance against the assumed 12% target.